

CDACL-002 – Customer Segmentation

Customer Segmentation and Shopping Trend Analysis Using SQL, Power BI and Orange Data Mining

Project Overview

This project analyzes the shopping behavior of **3,900 customers** from a U.S. retail shopping dataset. The main goal is to understand customer purchasing patterns, identify which products need discounts, study spending based on age, payment methods, seasons, and locations, and group customers into similar categories using **K-Means Clustering**. The results help businesses improve sales, customer satisfaction, and marketing strategies.

Objectives

- Analyze customer shopping behavior.
 - Find product categories where discounts should be applied.
 - Study spending based on age and payment method.
 - Analyze seasonal and location-wise sales.
 - Segment customers into **3 groups** using K-Means.
 - Recommend marketing campaigns for each customer group.
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Tools Used

- **MySQL** – Data Analysis using SQL
 - **Power BI** – Dashboard and Visualization
 - **Orange Data Mining** – Customer Segmentation
 - **Dataset (CSV)** – Shopping Data
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Simple SQL Analysis

SQL is used to answer business questions such as:

- Total Customers
- Total Sales
- Average Purchase Amount

- Sales by Category
- Age-wise Spending
- Payment Method Analysis
- Season-wise Sales
- Location-wise Sales
- Discount Analysis
- Subscription Status

These results are used to create Power BI dashboards.

Power BI Dashboard

The dashboard includes:

- Total Customers
- Total Revenue
- Average Purchase
- Sales by Category
- Sales by Season
- Sales by Location
- Payment Method
- Age-wise Spending
- Discount Applied
- Subscription Status

The dashboard helps managers quickly understand customer shopping trends.

Customer Segmentation (K-Means)

K-Means is a Machine Learning algorithm used to group customers with similar shopping behavior.

Features Used

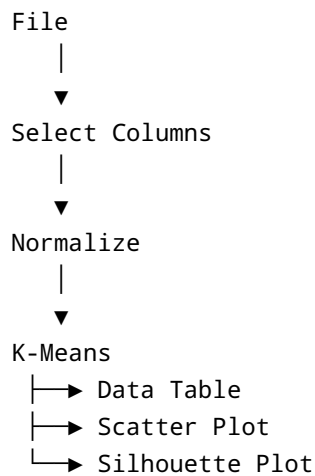
- Age
- Purchase Amount
- Previous Purchases
- Review Rating

How K-Means Works

1. Load the shopping dataset.
2. Select important customer features.
3. Normalize the data.
4. Choose **K = 3** clusters.

5. Randomly create 3 cluster centers (centroids).
 6. Assign each customer to the nearest cluster.
 7. Update the cluster centers.
 8. Repeat until the clusters no longer change.
 9. The final result is **3 customer groups**.
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Orange Data Mining Workflow



Customer Clusters

Cluster 1 – Budget Customers

- Low spending
- Use discounts frequently
- Buy low-cost products

Marketing: Coupons, flash sales, and bundle offers.

Cluster 2 – Premium Customers

- High spending
- Loyal customers
- Frequent purchases

Marketing: VIP membership, reward points, and personalized offers.

Cluster 3 – Seasonal Customers

- Shop mainly during festivals or special seasons
- Medium spending
- Use promotional offers

Marketing: Festival discounts, seasonal offers, and free shipping.

Dominant Factors

The main factors that influence customer segmentation are:

- Purchase Amount
 - Previous Purchases
 - Age
 - Review Rating
 - Season
 - Payment Method
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Business Benefits

- Understand customer behavior.
 - Identify valuable customers.
 - Improve marketing strategies.
 - Increase customer loyalty.
 - Optimize discounts.
 - Improve sales and business growth.
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Final Conclusion

This project successfully analyzes customer shopping behavior using **SQL**, visualizes insights with **Power BI**, and groups customers using **K-Means Clustering in Orange Data Mining**. Three customer segments—**Budget Customers, Premium Customers, and Seasonal Customers**—are identified based on shopping trends. The analysis helps businesses make better decisions, create personalized marketing campaigns, improve customer satisfaction, and increase overall sales.